

CONCEPT PAPER

Suggested Title

A Two-Layer Framework for API-Based Academic Credential Verification and Forgery Detection in Ethiopian Higher Education Institutions: A Benchmark Analysis of Machine Learning Techniques

1. Problem Statement

Academic credential forgery is becoming increasingly difficult to detect due to advances in digital editing and document manipulation technologies. Existing authentication approaches generally focus either on visual forgery detection using machine learning or on document integrity and record verification using techniques such as cryptographic hashing and blockchain. However, visual forgery detection alone cannot confirm whether credential information corresponds to an authentic institutional record, while integrity-based approaches may depend on previously registered documents or stored authentication information.

Therefore, there is a need for an approach that verifies both **credential information and visual document authenticity**. This study proposes a hybrid framework in which OCR extracts academic information for verification against authoritative institutional records through an API, while machine learning independently analyzes the document for visual forgery. A conflict-aware decision fusion mechanism will combine the record-match score and ML forgery score to produce the final authentication decision. The framework will be evaluated for its performance, reliability, simplicity, computational efficiency, and suitability for Ethiopian higher education institutions.

2. Objectives

General Objective

To design and evaluate a hybrid framework for authenticating Ethiopian academic credentials using OCR-API record verification, machine learning-based visual forgery detection, and decision fusion.

Specific Objectives

1. To develop an OCR-API mechanism for extracting and verifying academic credential information against authoritative institutional records.
2. To benchmark selected machine learning techniques for visual academic credential forgery detection.
3. To develop a conflict-aware decision fusion mechanism that combines record-match and ML forgery scores.
4. To compare record-verification-only, ML-only, and hybrid authentication approaches in terms of accuracy, reliability, simplicity, and computational efficiency.

3. High-Level Methodology

The study will follow a **Design Science Research (DSR) approach supported by experimental and comparative evaluation**. Recent Ethiopian and international studies will first be reviewed and compared based on their working procedures, methods, sample sizes, document authentication techniques, performance, simplicity, computational requirements, and limitations.

A localized dataset of genuine academic credentials will be prepared from selected Ethiopian higher education institutions, subject to institutional permission and ethical requirements. Representative forged samples will be produced through controlled manipulation of genuine documents to model relevant forgery types, including text alteration, signature and seal manipulation, template modification, and digitally or AI-generated forgery. The resulting dataset will be labelled and divided into appropriate training, validation, and testing sets.

The proposed framework will process each credential through **two complementary authentication paths**. In the first path, image preprocessing and Optical Character Recognition (OCR) will extract key fields such as student name, registration number, institution, degree/program, and graduation year. The extracted information will be submitted through an API and compared with authoritative institutional records. A verification module will calculate a **record-match score** based on the level of agreement between the extracted fields and the institutional record.

In the second path, the credential image will be independently analyzed for visual forgery. Selected machine learning techniques, including **CNN, ANN, SVM, Random Forest, and Logistic Regression**, will be trained and benchmarked under comparable experimental conditions. The selected model will produce an **ML forgery score** representing the likelihood that the submitted document has been visually manipulated.

The record-match score and ML forgery score will then be processed by a **conflict-aware decision fusion algorithm**. Agreement between the two verification paths will support the corresponding authentication decision, while conflicting evidence will be handled explicitly. For example, a high record-match score combined with a high ML forgery score may indicate that a genuine graduate record exists but the submitted certificate has been manipulated; such cases will be referred for manual review. Low OCR confidence, poor document quality, or unavailable API verification will also be treated as uncertainty conditions. Final outcomes will include **Verified Genuine, Suspected Forged, Manual Review Required, or Unable to Verify**.

The ML techniques will be evaluated using **Accuracy, Precision, Recall, F1-Score, AUC-ROC, MCC, false-positive and false-negative rates, processing time, and computational requirements**. OCR extraction and record-verification performance will also be assessed. Finally, **record-verification-only, ML-only, and hybrid fusion approaches** will be compared to determine whether the proposed framework improves authentication reliability while maintaining acceptable simplicity and computational efficiency for Ethiopian higher education institutions.

Expected Contribution

The study is expected to contribute an empirically evaluated hybrid academic credential authentication framework that combines **authoritative institutional record verification with visual forgery detection**.

Its main contribution will be the integration and comparative evaluation of these complementary authentication mechanisms through a **conflict-aware decision fusion algorithm**, together with benchmark evidence for selecting a suitable machine learning technique for Ethiopian higher education institutions.